University of Nottingham
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Applications of AI in EEE

EEEE3134

Zhengbing He

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Team

- **Zhengbing He**
  - Professor in *Autonomous and Intelligent Systems*
  - My research lies at the intersection of urban mobility, systems engineering, and artificial intelligence,
    - spanning from traditional topics such as traffic flow operations and control, sustainability, and resilience to emerging areas including data-driven modeling, autonomous driving, and large language models
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About the Course

- **Student Summary of Content**
  - This module provides an introduction to **AI for engineers who are curious about AI.**
  - The module considers what AI is, what it is used for and why it is used in engineering.
  - The module **avoids the mathematics** of the subject and instead focusses on the concepts and application of AI in the context of AI as an engineering tool.
  - The module introduces the core concepts of AI (such as what is meant by AI, machine learning and deep learning), supervised vs unsupervised learning and examples in engineering.
- **Assessment**

	Weight	Type	Duration	Requirement
Coursework	40	Report		Construct, validate and demonstrate an AI model and provide brief report up to 6 pages, on model metrics.
Exam	60	in Person	2h	ExamSys

About the Course

- **Syllabus/Structure**

Artificial Intelligence

- Concepts and Historical Development (week 3)

OVERVIEW

- Large Language Models
- Convolutional Neural Network
- Recurrent Neural Network (optional)
- Reinforcement Learning (optional)

(weeks 7/8-12/13)

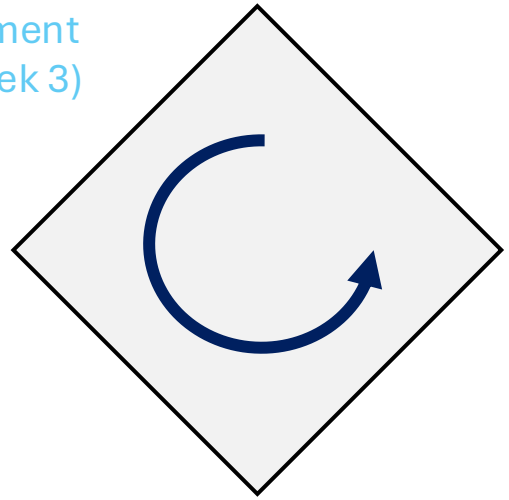
The Landscape of AI Technologies

What Does Machine Learning Learn

SKILLS

Modern Programming

- Python, VS Code, Git/GitHub, and AI Agent (week 4)



TOOLS

APPLICATIONS

Autonomous Driving

- Concepts and Historical Development
- Vehicle Longitudinal Motion Modeling
- Gipps Model
- kNN-Driven Model

Coursework (40 pts): 04-Nov to 05-Dec (weeks 5-6/7)

Thank You

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